1. Explain the role of the Indus Valley Civilisation in early mathematics.

2. Write down the major differences between mathematics in the Indus Valley Civilization and the Vedic Period?

3. What is the contribution of Pāṇini to Indian intellectual thought?

4. What does "Śulba" mean? Mention the name of four Sulbakars and their contributions in Sulbasutras? Explain the role of the Sulba Sutras in early Indian mathematics.

5. Geometrically proof $(a-b)^2 + 2ab = (a)^2 + (b)^2$

6. How a circle can be constructed from square according to Baudhayana?

7. How a square can be constructed from a rectangle?

8. What sources are used to reconstruct Indus Valley mathematics?

9. What geometric features are visible in Indus town planning?

10. Mention two examples of geometry from the Indus Valley.

11. Did Indus Mathematics Influence Later India?

12. Explain the deicmal binary system in Indus Valley Civilisation in early mathematics with examples.

13. Identify the number represented by the Katapayadi system in the following cases

a) Līlāvatī

b) Anuyogadvārasūtra

c) Sulbasutras

d) Brāhmasphuṭasiddhānta

e) Anandamrtam

f) Rakshabandhanotsavah

g) Korona Rajanaskah

h) Aryabhatiya

i) Chandahṣāstra

j) Ganita-Sara-Sangraha

k) Siddhanta Shiromani

l) Sūryaprajñapti

Katapayadi System / कटपयादि

कटपयादि संख्या - kaTapayAdi for Melakarta Ragam Names & Numbers									
1	2	3	4	5	6	7	8	9	0
क	ख	ग	घ	ङ	च	छ	ज	झ	ञ
ट	ठ	ड	ढ	ण	त	थ	द	ध	न
प	फ	ब	भ	म					
य	र	ल	व	श	ष	स	ह		
ka	kha	ga	gha	nga	cha	Cha	ja	Jha	nya
Ta	Tha	Da	Dha	Na	ta	tha	da	dha	na
pa	pha	ba	bha	ma					
ya	ra	la	va	sha	Sha	sa	ha		

14. Identify the number represented by the Bhuta-sankhya system in the following cases:

a) rama-chandra-guna-nanda-risi-vedah

b) guna-adri-kha-sastra-nayana-bhujangah

c) purna-tri-veda-gruha-nanda-rtu-padah

Write above numbers using Katapayadi system.

15. Rewrite the numbers below using the bhuta-samkhya system using Table 6.3 below.

a) 179,532,695

b) 345, 447

c) 670,087,123

TABLE 6.3 Word Numerals for Bhūta-saṃkhyā System

Number	*Represented by (partial list only)**
0	śūnya, anata, pūrṇa, kha
1	ādi, candra, prithivī, eka
2	aśvin, pairs of limbs, ayana, dvandva, dvi
3	rāma, guṇa, loka, kāla, agni, trinetra
4	veda, śruti, yuga, āśrama, varṇa, samudra, kṛta
5	bhūta, śāstra, bāṇa, pāṇḍava, indriya
6	aṅga, ṛtu, darśana, ṣaṇmukha, ṣaṭ
7	ṛṣi, adri, svara, dhātu, chandas
8	vasu, bhujaṅga, siddhi, dik, kuñjara, nāga
9	gṛha, aṅka, nanda
10	dik, aṅgulī, avatāra, rāvaṇaśiras
11	rudra
12	āditya, rāśi
13	viśva, kāma
14	manu
15	tithi, dina

* For all the items listed, any word from the synonyms may be used

Write above numbers using Bhuta-sankhya system.

16. Sayanacharya gives us the speed of sunlight in ½ nimesa is 2202 yojanas. Calculate speed of the light? How accurate is this value with modern value.

17. Using Vedic Sutras solve the followings :

a) 'Paraavartya Yojayet' : 36520981 ÷ 133

b) 'Sopaantyadvayamantyam' : 624 X 12 = ?

c) 'Yaavadunam' : 989 X 1005 =?

d) 'Dhvajanka' : 601325 ÷ 76 = ?

18. What is cyclic quadrilateral? Brahmagupta gives the following property of the diagonals of a cyclic quadrilateral:

The sums of the products of the sides about each diagonal are divided by one another and multiplied by the sum of the products of the opposite sides. Taking square roots gives the diagonals in a viṣa ma (unequal) quadrilateral.

If a = 39, b = 52, c = 60, d = 25 are the lengths of sides of a cyclic quadrilateral then find its diagonals x, y based on the above property. Find also the area and circumradius of this cyclic quadrilateral.